PAUL KNOPF

Lead Engineer/Architect

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CAREER SUMMARY

Passionate engineer and problem solver — doing what I love for a living.

I have a deep understanding of the SDLC and strong leadership skills to match. I've led and scaled large teams building mission-critical systems, establishing architecture, processes, and best practices to keep things reliable and moving.

I'm a big believer in simplicity and low ceremony. My solutions are built to last, move fast with changing requirements, and stay accessible enough that even junior engineers can contribute effectively.

Strong advocate for AI with a proven track record in production deployments, observability and team management.

As an IC, I'm full-stack across the board: infrastructure provisioning, security/networking, application development, testing, deployment, and observability. Always learning and staying current.

I also maintain a few open-source projects across JavaScript, C#, and Go.

SKILLS & TOOLS

General

Architecture/Design · AI/Agentic Coding · Networking/Security · Mentoring/Speaking · Agile/Scrum

Languages

C# · HTML/CSS · TypeScript/JavaScript/Node · Go · C++/C · Unix Shell Scripts · SQL · Python · Terraform

Client-side

React · Redux/MobX · webpack/vite · Jest · Sass/LESS · Bootstrap · jQuery

Mobile

iOS/Android (Xamarin) · Xcode · UIKit · Objective-C

Operating Systems

Unix/Linux · macOS · Windows

Cloud

Terraform · Microsoft Azure · Google Cloud · Kubernetes · Containers (Docker/LXC) · Azure Pipelines/GitHub Actions/Jenkins

Databases

Postgres · MongoDB · SQL Server · SQLite · Cassandra · Lucene/ElasticSearch

LETTERS OF RECOMMENDATIONS

...our company would not have achieved the level of technical maturity and product success that it did without his contributions. He played a significant role in building the foundation that positioned Med X Change for acquisition, and I remain grateful for the impact he had on our organization...

...Paul was someone I trusted to tackle our most challenging initiatives. He consistently demonstrated ownership, sound judgment, and an unwavering commitment to delivering results...

...I recommend Paul without reservation. Any organization seeking an exceptional engineering leader, software architect, or technical executive would be fortunate to have him on their team. I am confident he will bring the same level of integrity, innovation, and execution that made him invaluable at Med X Change...

Seth Kardos *CEO, Med X Change* Read more

...enthusiastically recommend... for any role as a Software Engineer Lead or Architect...

...one of the brightest engineers I have ever worked with...

...his technical skills are exceptional, and he consistently demonstrated a deep understanding of software architecture and engineering principles...

...his personal skills, including strong communication and teamwork, make him a pleasure to work with.

Rodrigo Bercocano do Amaral *Software Engineer Sr Manager, Ensemble Health Partners* Read more

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...I've consistently seen him demonstrate strong technical capability, ownership, and a thoughtful approach to complex engineering challenges...

...Overall, Paul is a dependable and capable engineer who contributes meaningfully to both the design and evolution of complex systems. He works well across teams, takes ownership of important areas, and consistently adds value in technical discussions and implementation efforts...

...I would strongly recommend him for roles that require solid engineering fundamentals, collaborative problem-solving, and the ability to contribute to shared platform infrastructure...

Scott Long *Lead Software Engineer, Ensemble Health Partners* Read more

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...I was continuously impressed by his breadth of knowledge and ability to learn and gain expertise in a wide array of technologies...

...Frequently Paul would bring to the team a new technology or way of doing things that later proved to be a very good idea. Rather than just scratch the surface he would dig in and become extremely knowledgeable on technology that the team was using. I believe Paul did a good job at choosing technology that was the right fit for the task. Sometimes that meant using a new technology and sometimes it meant knowing when to use something that had been proven over time already...

...confident he would be a benefit to any software team where he is able to lead and contribute to the technical architecture of a solution or product...

Connor Flaat *Principal Software Engineer, TTEC Digital* Read more

WORK EXPERIENCE

Principal Solutions Architect

Allied Benefit Systems

Aug 2026 - Present

Lead Software Engineer

Ensemble Health Partners

Aug 2022 - June 2026

- Led several teams and initiatives across AI, infrastructure, and platform engineering.
- Azure AI Foundry deployment — pool of models behind a single load-balanced endpoint (OpenAI API).
- Datalake Lakebase Postgres for integration to Unity Catalog.
- Databricks LangChain agents, scores/evaluations, for patient/account resolutions.
- Designed and rolled out an agent-to-agent protocol for a distributed pool of specialized agents.
    - Temporal for long-running/durable sessions with traceability and monitoring.
    - Orchestrator agent that discovers sub-agents and their tools/schemas.
    - Base ASP.NET libraries for teams to expose high-fidelity API endpoints as discoverable tools.
- Curated shared Claude Code skills to enforce consistent usage of languages and tools (.NET, JS, Terraform).
- Claude agent that monitors Application Insights (webhooks) for exceptions and auto-creates Azure DevOps issues — capturing stack traces, relevant logs, a breakdown of the exception, and possible fixes.
- Pipeline routines for running evaluations against LLM prompts to catch regressions and validate accuracy.
- Terraform modules for defining multi-region, multi-env resources.
    - APIM/WAF
    - Hub/Spoke
    - Team definitions for dedicated Azure Subscriptions and spokes (for hub).
- Rolled out Kubernetes clusters and established the process for teams to deploy, maintain, and monitor their workloads (Flux).
- Designed and rolled out a micro-frontend architecture, letting teams develop modules locally and deploy them into a single integrated client-side app.
- Designed and rolled out CQRS pattern — addressed upper performance limits of the existing architecture, resulting in dramatic performance gains and significant cost savings.
- Regular company-wide presentations on new technologies, best practices, and architecture.

Technologies used:

Azure AI Foundry (OpenAI models) · Claude Code · .NET Core · Microservices · Dapper/OrmLite · Kubernetes · React/TypeScript · Micro-frontends · Temporal · Azure DevOps · Azure · Terraform · ElasticSearch · Postgres · Redis

Technical Lead

Novanta Surgical Imaging

July 2019 - Aug 2022

- Designed and implemented next-gen prescriptive architecture for medical imaging software.
- Protobuf/gRPC microservice architecture with a focus on performance and stability (emphasis on Go).
- Coordinated with the firmware team to keep driver and software development in feature parity.
- ASP.NET/React multi-tenant application with on-site deployments and automated updates.
- Established usage of IDesign's "The Method" for system and project design.

Technologies used:

.NET · ASP.NET MVC · Microservices · Protobuf/gRPC · Docker · SQLite/MongoDB/Postgres · C/C++ · Golang · JavaScript/TypeScript · Microsoft Azure · CI/CD (Pipelines) · Unit Testing

Senior Software Engineer

Med X Change

Nov 2012 - July 2019

- Thought leader for a small embedded and web development team.
- Managed IT infrastructure: build agents, on-prem GitLab, project management tooling.
- Embedded .NET Core runtime in a native executable with V8 interop.
- iOS/Android development using Xamarin with ASP.NET WebAPIs for mobile consumption.
- DirectShow and GStreamer for encoding/decoding various video formats (H264/MP4).
- WPF and Qt/QtQuick digital recording software on embedded devices.
- C/C++ libraries for video capture and hospital network integration (DICOM/HL7).
- Custom embedded Linux distros with Yocto — brickless updates, read-only boot with tmpfs overlay for stability and resilience.
- Cross-compilation across vc++, gcc, MAKE, CMAKE, meson, and autotools.

Technologies used:

.NET · ASP.NET MVC · WPF · Qt · Docker · SQLite/Postgres · C/C++ · Golang · JavaScript/TypeScript · DICOM/HL7 · Linux (Embedded, Sys Admin) · Google Cloud Platform · CI/CD (Pipelines) · Unit/Integration Testing

Software Engineer

MethodFactory

Jun 2011 - Dec 2012

- Managed production IIS/SQL Server instances for several high-volume sites (coleman.com, rawlings.com).
- Maintained a prescriptive C# .NET 4.0 framework used to accelerate application development to industry best practices.
- Managed an internal automated NuGet feed (via TFS Build) for delivering framework updates.
- Database design, index/performance tuning, and troubleshooting with SQL Profiler.
- Built a process for managing SQL Server Database Projects (SSDT) — distributing .dacpac files alongside framework libraries for automated schema deployment and CI validation.

- Implemented an enterprise WCF Service Host Factory using IoC/DI for runtime composability.
- Built an MVC3/jQuery eCommerce UI foundation delivered as a VirtualPathProvider zip, with full CSS/JS/view override support via a custom ViewEngine and Cassette for client-side dependency management.
- Designed an N-tier enterprise framework reused across multiple clients (Coleman, Nuk, Rawlings).
- Built medical equipment software for image/video capture using MSMQ/NServiceBus/SignalR for hardware-to-web messaging.

Technologies used:

.NET | ASP.NET MVC | SQL Server | TFS | Microsoft Test Manager | Microsoft Commerce Server | Windows Server 2008/2012 | Microsoft BizTalk

Freelance Software Engineer

Adept Web Studios

Jun 2009 - June 2011

Software Developer

Dataline Associates

Sep 2009 - Aug 2011

PROJECTS

Qml.Net github.com/qmlnet/qmlnet

.NET and QML integration for cross-platform, high-performance desktop apps. Built a thin C layer using C++ shared pointers to bridge the two garbage collectors (.NET and QML/JavaScript). CI configured with AppVeyor and Travis CI across Linux, macOS, and Windows.

Technologies used:

C/C++ .NET Core Qt Continuous Integration

Darch github.com/godarch/darch

Think "Dockerfiles" but for immutable, stateless environments booted bare-metal. Each distro uses a custom initramfs that overlays a tmpfs over read-only images — every boot is a clean boot. Runtime changes (apt-get, etc) are discarded on reboot; anything you want persisted has to be baked into a recipe. I use it as my daily driver.

Technologies used:

Go Docker containerd Continuous Integration

EDUCATION

Associates

SCF